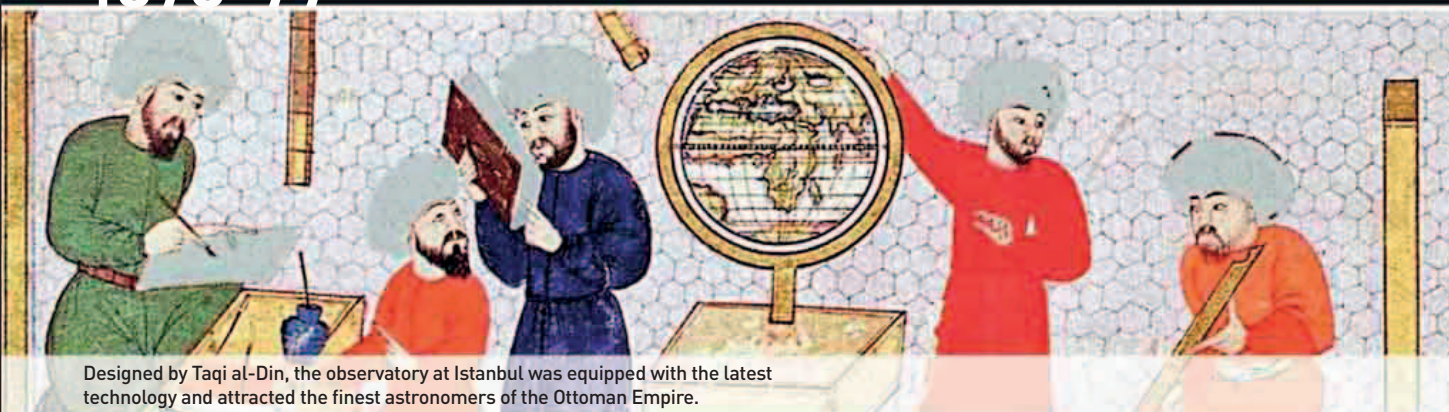




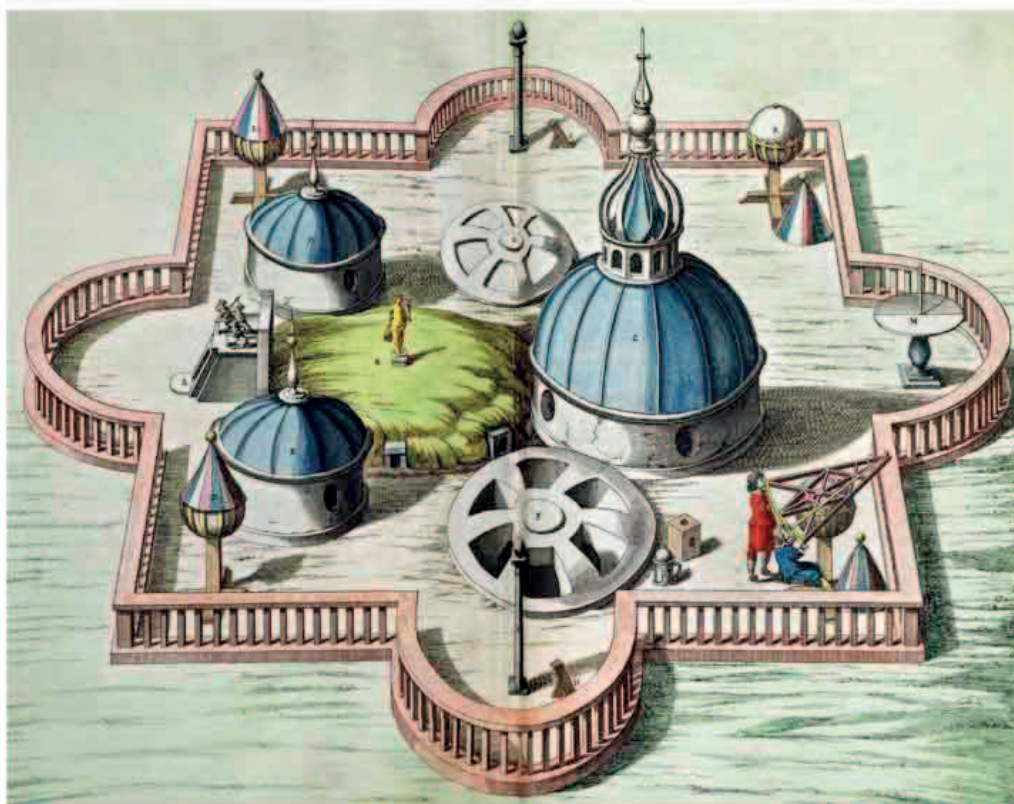
Designed by Taqi al-Din, the observatory at Istanbul was equipped with the latest technology and attracted the finest astronomers of the Ottoman Empire.

AN INHABITANT OF SICILY,

Greek mathematician and astronomer **Francesco Maurolico** (1494–1575) published several treatises on mathematics. In *Arithmeticonum Libri Duo* (*Two Books on Arithmetic*), published in 1575, he was the first mathematician known to prove a mathematical statement explicitly using **mathematical induction**. This is a method of proof using a series of successive logical steps.

To persuade astronomer **Tycho Brahe** (see 1572–74) to return to his native Denmark, King Frederick II offered him land and funding to **establish an observatory** in Hven (an island now belonging to Sweden). Work on the building, known as **Uraniborg**, began in 1576. However, the completed structure was not considered steady enough for accurate observations. A second complex, **Stjerneborg**, was built close by in 1584 to house the delicate equipment. Together, these two complexes formed a **major centre for astronomical and scientific research**.

Italian polymath **Gerolamo Cardano** trained in medicine and was a respected physician. He wrote several treatises, including the **first description of typhoid fever** in 1576. He was the first to recognize the disease's distinctive symptoms.



There were **many advances in botany** in the second half of the 16th century. Emphasis moved from the study of plants for their medicinal properties to a more comprehensive study and classification of plant life. Botanist **Charles de l'Écluse** (1526–1609), also known as Carolus Clusius, published the first of his studies of the flora of Spain in 1576. He went on

to found the botanical garden at the University of Leiden in Holland, where his work helped lay the foundation of the Dutch tulip industry.

While Brahe was sponsored by the King of Denmark, the Ottoman Turkish engineer and astronomer **Taqi al-Din** persuaded Sultan Murad III to fund an equally prestigious observatory in Istanbul. Built

Stjerneborg observatory

The complex of buildings that replaced Tycho Brahe's Uraniborg observatory was equipped with the latest astronomical instruments.

in 1577, this was **designed to be the major observatory of the Islamic world**. However, it existed only briefly: after a mistaken astrological prediction of Ottoman victories in battle, the Sultan had the observatory destroyed in 1580.



IN 1579, HIERONYMUS FABRICIUS

(1537–1619), professor of anatomy and surgery at the University of Padua, noticed in his dissections folds of tissue on the inside of veins. He described these folds as valves, but did not propose any function for them. It was only later that they were found to prevent the backflow of blood as it returns to the heart. Fabricius's treatise on the subject, *De Venarum Ostioliis* (*On the Valves of the Veins*), particularly influenced one of his later students, William Harvey (see 1628–30).

Although trained in medicine, Venetian physician **Prospero Alpini** (1553–c.1616) was more interested in botany. In 1580, he took up a post as physician to the Venetian consul in Cairo, Egypt, where he studied the plant life. He also worked as the manager of a date palm plantation in Egypt. While there, he observed that the **pollination of flowers was necessary to produce fruit**, and deduced that there were two sexes of plants. Alpini's study of plants in Egypt inspired him to write several books on exotic plants, including *De Plantis Aegypti Liber* (*Book of Egyptian Plants*), published in Venice in 1592, and *De Plantis Exoticis* (*Of Exotic Plants*), published posthumously in 1629. He is also credited with introducing the banana and baobab to Europe.

1575 Francesco Maurolico makes the first known use of mathematical induction

1576 Tycho Brahe builds an observatory called Uraniborg on the island of Hven, Denmark

1577 Taqi al-Din builds an astronomical observatory in Galata, Istanbul

1579 Hieronymus Fabricius describes valves in veins

1576 Gerolamo Cardano gives the first description of typhoid fever

1576 Carolus Clusius publishes the first of his books on botany